

PRAKASH MAHESHWARAN

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EDUCATION

Binghamton University, SUNY – M.S. Computer Science | Expected March 2026 | GPA 3.74/4.00 | Clark Fellowship

Relevant Coursework: Machine Learning, Artificial Intelligence, Data Mining, Introduction to Artificial Intelligence

VIT ICER, Bangalore – Postgraduate CS (AI Track) | March 2024 | GPA 3.80/4.00

Relevant Coursework: Machine Learning, Artificial Intelligence, Deep Learning, Python for Data Analytics

CAHCET, Anna University – B.Tech. Information Technology | March 2023 | GPA 3.30/4.00

Relevant Coursework: Problem Solving and Python Programming, Engineering Mathematics I, Probability and Statistics, Object-Oriented Programming

TECHNICAL SKILLS

Languages: Python, C++, Java, MATLAB, R

Specialized Skills: Machine Learning, Deep Learning, PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Circuit Simulation, SPICE, HSPICE

Technologies: Bayesian Optimization, Evolutionary Search, Reinforcement Learning, Physics-Informed ML, GNNs, Surrogate Models, Data Pipelines, Active Learning, Uncertainty Quantification, Multi-objective Optimization

Other Expertise: Cadence Virtuoso, ADE, Layout Extraction, Parasitic Extraction, OLED, Gate Driver Design, Hardware Bring-up, Analog Circuits, Digital Circuits, EDA Tools

Certifications: Introduction to Machine Learning, Python for Data Science, Deep Learning Specialization

PROFESSIONAL EXPERIENCE

Senior Research Assistant, SUNY RF | New York, USA | March 2025 – Present

- Developed ML models to predict electrical KPIs from layout parameters, reducing simulation time by 30%.
- Implemented Bayesian optimization for multi-objective circuit design, generating 15% more efficient candidate topologies.
- Collaborated with senior engineers to integrate RL agents for process control parameter tuning, improving yield by 5%.

Junior Intern, Skyvern (Y Combinator) | May 2025 – Aug 2025

- Built and validated surrogate models for parasitic extraction, accelerating simulation loops by 50%.
- Engineered data pipelines for fab and test data, enabling active learning strategies for improved model accuracy.
- Applied uncertainty quantification techniques to surrogate models, providing confidence intervals for predictions.

PROJECT EXPERIENCE

Physics-Informed Neural Networks for Parasitic Extraction | Chennai, India | Jul 2023 – Feb 2024

- Problem:** Traditional parasitic extraction is computationally expensive and limits design space exploration.
- Approach:** Developed PINNs incorporating physical device equations and layout features to create a fast, accurate surrogate model.
- Impact:** Achieved over 40% reduction in simulation time while maintaining high accuracy, enabling rapid what-if analysis for circuit designers.
- Outcome:** Demonstrated the feasibility of differentiable simulation for end-to-end AI-driven circuit design optimization.

Reinforcement Learning for Adaptive Process Control | Chennai, India | Mar 2023 – Jul 2024

- Problem:** Maintaining optimal process parameters in semiconductor fabrication is challenging due to variability.
- Approach:** Designed an RL agent trained on simulation data to dynamically adjust process control parameters for improved yield.
- Impact:** The RL agent demonstrated a 7% improvement in wafer yield compared to traditional PID controllers in simulated environments.
- Outcome:** Provided a framework for intelligent, adaptive control systems in advanced manufacturing processes.

Graph Neural Networks for Layout-to-Electrical Performance Prediction | New York, USA | May 2023 – Oct 2024

- Problem:** Predicting electrical performance from complex circuit layouts is a difficult task for conventional ML models.
- Approach:** Leveraged GNNs to capture spatial relationships and dependencies within circuit layouts for accurate electrical KPI prediction.
- Impact:** The GNN model achieved R-squared values above 0.95 for critical parameters like delay and power consumption.
- Outcome:** Enabled faster design iterations and more reliable performance predictions early in the design cycle.

WHITE PAPERS

AI-Accelerated Circuit Design: Bridging Physics and Machine Learning | Int'l Journal of AI Research | August 2025

- Explores the synergistic integration of physics-based simulations with advanced ML techniques like GNNs and PINNs.
- Discusses the potential of Bayesian optimization and RL in optimizing complex design spaces and controlling fabrication processes.
- Highlights the development of high-fidelity surrogate models for parasitic extraction and performance prediction.

Surrogate Modeling for Rapid Circuit Simulation and Optimization | Preprint Archive | June 2025

- Presents methodologies for building efficient surrogate models using ML to approximate complex circuit simulations.
- Details techniques for uncertainty quantification and active learning to improve surrogate model accuracy and robustness.
- Proposes a framework for combining surrogate models with optimization algorithms for accelerated design loops.

LEADERSHIP EXPERIENCE

Event Coordinator, Students Guidance Cell | Chennai, India | 28 Months

Organized over 15 student development events, improving engagement and support services for 500+ students.

Volunteer Teacher (Math & CS), U&I | Vellore, India | 12 Months

Taught math and computer science to underprivileged students, improving academic performance for over 30 students.

ACHIEVEMENTS AND AWARDS

- Winner, National Semiconductor Design Competition, IIT Madras
- Presented Research on AI in EDA at IEEE Conference, Bangalore
- First Place, University Hackathon for AI-driven Optimization, VIT Chennai
- Published Paper on GNNs for Circuit Layout Analysis in Journal of AI Research